DIRECTORY

During World War II, governments increased funding and support for research, hoping to gain tactical advantages. Intense competition among scientists racing to make new findings often led to valuable discoveries for civilians, as well as the military, and continued after the war was over.

WOLFGANG PAULI

1900–1958

Austrian-born Swiss and American theoretical physicist Wolfgang Pauli is best known for his work on atomic structure. He created the Pauli exclusion principle, which states that no two electrons in an atom can exist in exactly the same state at the same time. Pauli (and, independently, German theoretical physicist Arnold Sommerfeld) devised an atomic model that explained the electrical and thermal properties of metals. Pauli was also the first to propose the existence of the subatomic particle the neutrino. He was awarded the Nobel Prize in Physics in 1945.

THEODOSIUS DOBZHANSKY

1900–1975

Russian population geneticist Theodosius Dobzhansky was a central figure in the field of evolutionary biology. Born in Ukraine, he emigrated to the US in 1927. Largely through experiments with fruit flies, he helped to bridge the gap between genetics and evolution. He showed that mutation, in creating genetic diversity, supplied the raw material for natural selection to act on. In this way, he helped reconcile the findings of genetics with Charles Darwin's theory of evolution, which placed emphasis on natural selection.

HOWARD AIKEN

1900–1973

American electrical engineer Howard Aiken was responsible for building the Harvard Mark I computer in 1944. Developed for use in World War II, this part-electric, part-mechanical computer showed that programmable machines could crack large-scale problems systematically, reliably, and automatically; it helped to set the stage for today's computers. Aiken worked closely with Grace Hopper, one of the originators of computer programming.

WERNER HEISENBERG

1901–1976

Born in Würzburg, southern Germany, in 1901, Werner Heisenberg studied mathematics and physics at the universities of Munich and Göttingen, meeting his future collaborator Niels Bohr in Göttingen in 1922. Heisenberg is best known for his work on the Copenhagen interpretation (which looks at the way quantum systems, governed by the laws of probability, interact with the large-scale world) and the uncertainty principle (which says only the speed or the position of a particle is known, not both). He also contributed to quantum field theory and worked out his own theory of antimatter. At age 32, he was awarded the Nobel Prize in Physics, making him one of its youngest recipients.

LINUS PAULING

1901–1994

Chemist Linus Carl Pauling was born in Portland, Oregon, and was the winner of two Nobel Prizes—one for Chemistry and the other for Peace (awarded for his attempts at mediating between the US and Vietnam). Pauling was an early proponent of quantum mechanics in chemistry, and his book *The Nature of the Chemical Bond*, published in 1939, is one of the most influential chemistry texts ever published. He set out five rules for finding the structure of ionic crystals and described how atomic orbitals in covalent compounds combine to form molecular orbitals. According to Pauling, when two atoms form a covalent bond, they share electrons but their position is not fixed; instead, the electrons can move to a more favorable position to bond.

KATHLEEN LONSDALE

1903–1971

Working under the direction of William Henry Bragg at the Royal Institution, London, Kathleen Lonsdale developed